

$$P_{\max} \leq 60$$

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

Recent work established the structural bounds

$$N_{\max} = 4,$$

$$C_{\max} = 5,$$

and

$$B_{\max} = 128,$$

representing the maximum numbers of sovereign nations, currencies, and banking institutions within a unified monetary architecture.

The present paper introduces a mathematical definition of a political party as an economic institution rather than merely an electoral organisation.

Each political party is assumed to

1. possess exactly one domestic currency,
2. own shares in one or more central banks,
3. participate in the unique international market portfolio.

Assuming one central bank per nation and recognising that the international market portfolio exists only because of the fifth currency, a purely combinatorial counting argument yields

$$P_{\max} \leq 60.$$

The proof illustrates how political structures may be interpreted as ownership configurations over monetary institutions. The resulting framework provides a bridge between political economy, monetary economics, institutional finance, graph theory, and combinatorics.

The paper ends with “The End”

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1 Introduction

The architecture of modern civilisation consists of numerous interacting institutional systems. Among the most fundamental are

- sovereign nations,
- monetary systems,
- banking systems,
- political organisations,
- international financial markets.

Previous papers established several structural upper bounds.
The first theorem showed that

$$N_{\max} = 4,$$

meaning that no more than four sovereign nations can coexist within the proposed monetary architecture.
The second theorem demonstrated

$$C_{\max} = 5,$$

placing an upper bound on the number of currencies.
The fifth currency represents the international trading currency required to support the international market portfolio.
The third theorem established

$$B_{\max} = 128,$$

giving an upper bound on the number of banking institutions.
The present paper addresses a natural remaining question.

How many political parties can exist?

Unlike traditional political science, our definition of a political party is economic.
Political parties are interpreted as ownership structures over monetary institutions.
Consequently, their number becomes a combinatorial quantity rather than merely an empirical observation.

2 Definitions

Definition 1. *A political party is defined as an institution possessing*

1. *one domestic currency,*
2. *ownership of shares in at least one central bank,*
3. *ownership of the international market portfolio.*

Definition 2. *A domestic currency is the legal currency issued by one of the sovereign nations.*

Definition 3. *The international currency is the unique currency supporting the international market portfolio.*

Definition 4. *A central bank is the unique monetary authority associated with one sovereign nation.*

3 Architectural Assumptions

Throughout the paper we assume

$$N_{\max} = 4,$$

and therefore

$$\text{Number of central banks} = 4.$$

Similarly,

$$C_{\max} = 5.$$

The five currencies consist of

1. Nation 1 currency,
2. Nation 2 currency,
3. Nation 3 currency,
4. Nation 4 currency,
5. International currency.

The fifth currency is fundamentally different from the first four.

Rather than serving as a domestic monetary unit, it enables international portfolio ownership.

4 Economic Interpretation

Traditional political parties are usually classified according to

- ideology,
- electoral support,
- historical evolution,
- constitutional law.

The present framework instead characterises political parties by their balance sheets.

Each party becomes an institutional investor possessing

- monetary assets,
- equity in central banks,
- international financial assets.

This interpretation naturally links political economy with modern asset pricing and institutional finance.

5 Notation

Let

$$P = (c, S, M),$$

where

- c denotes the domestic currency,
- S denotes the set of central banks in which shares are held,
- M denotes ownership of the international market portfolio.

The domestic currency satisfies

$$c \in \{1, 2, 3, 4\}.$$

The central-bank ownership set satisfies

$$S \subseteq \{1, 2, 3, 4\},$$

with

$$S \neq \emptyset.$$

The international market portfolio is unique.
Therefore,

$$M = 1.$$

6 Combinatorial Structure

There are

$$4$$

possible domestic currencies.

The number of non-empty subsets of four central banks equals

$$2^4 - 1.$$

Since

$$2^4 = 16,$$

we obtain

$$2^4 - 1 = 15.$$

Finally,
the international market portfolio contributes only 1 possible state,
because every political party is required to possess it.
Hence

$$1.$$

The complete counting argument will be presented in this paper.

7 The Injectivity Axiom

The preceding definitions describe the economic composition of a political party. To convert the combinatorial enumeration of ownership structures into an upper bound on the number of political parties, one additional modelling assumption is required.

Definition 5 (Ownership Configuration). *The ownership configuration of a political party is the ordered triple*

$$\Pi = (c, S, M),$$

where

- c denotes the unique domestic currency,
- S denotes the non-empty set of central banks in which the party owns shares,
- M denotes ownership of the international market portfolio.

Definition 6 (Injectivity Axiom). *Distinct political parties possess distinct ownership configurations.*
Equivalently,

$$\Pi_i = \Pi_j \implies i = j.$$

Therefore,

$$i \neq j \implies \Pi_i \neq \Pi_j.$$

The Injectivity Axiom states that the ownership configuration constitutes a complete identifier for a political party. Consequently, two parties cannot possess identical combinations of

1. domestic currency,
2. central-bank ownership,
3. international market portfolio.

The assumption transforms the counting of ownership configurations into the counting of political parties.

8 Preliminary Lemmas

The proof of the main theorem follows from three elementary counting lemmas.

Lemma 1. *There exist exactly four admissible domestic currencies.*

Proof. By assumption,

$$N_{\max} = 4.$$

Each sovereign nation issues exactly one domestic currency.
Therefore,

$$|\mathcal{C}_D| = 4.$$

□

Lemma 2. *There exist exactly four central banks.*

Proof. Each sovereign nation possesses one central bank.

Since

$$N_{\max} = 4,$$

the total number of central banks equals

$$4.$$

□

Lemma 3. *The number of admissible central-bank ownership structures equals*

$$2^4 - 1 = 15.$$

Proof. Let

$$\mathcal{B} = \{B_1, B_2, B_3, B_4\}$$

denote the set of central banks.

A political party may own shares in any non-empty subset of $\mathcal{B}$.

The power set of $\mathcal{B}$ contains

$$2^4$$

subsets.

Removing the empty set leaves

$$2^4 - 1 = 15$$

admissible ownership structures.

□

Lemma 4. *The international market portfolio contributes exactly one ownership state.*

Proof. The fifth currency is the unique international currency.

The international market portfolio is defined relative to this currency.

Every political party is required to possess ownership of the international market portfolio.

Hence there exists only one admissible state,

$$M = 1.$$

Therefore the international market portfolio contributes the multiplicative factor

$$1.$$

□

9 Enumeration of Ownership Configurations

The previous lemmas imply that the number of admissible ownership configurations equals

$$4 \times 15 \times 1.$$

More explicitly,

Component	Number of Choices
Domestic currency	4
Central-bank ownership	15
International market portfolio	1
Total	60

Table 1: Enumeration of ownership configurations.

Hence the ownership space contains

$$60$$

distinct elements.

10 Graph-Theoretic Representation

The ownership architecture may be represented by a labelled graph.

Let

$$G = (V, E),$$

where

$$V = \{N_1, N_2, N_3, N_4, B_1, B_2, B_3, B_4, I\},$$

with

- N_i denoting sovereign nations,
- B_i denoting central banks,
- I denoting the international market portfolio.

Each political party corresponds to

1. exactly one nation vertex,
2. one or more central-bank vertices,
3. the unique international-market vertex.

Consequently,

each ownership configuration is a labelled subgraph satisfying the preceding incidence constraints.

11 Illustrative Architecture

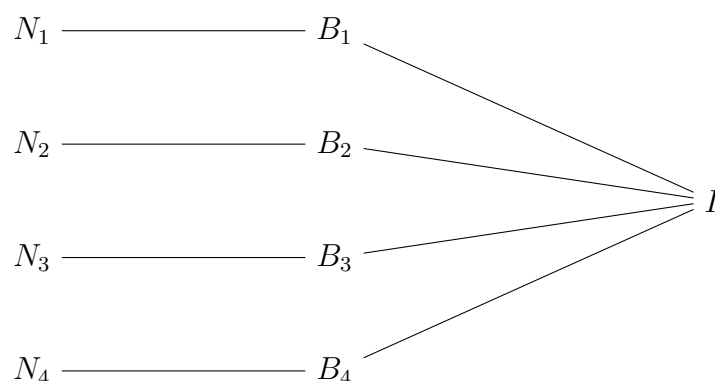


Figure 1: The four-nation, four-central-bank and unique international market portfolio architecture.

Every admissible political party selects one nation, a non-empty subset of central banks and the unique international portfolio.

The combinatorial enumeration is now complete.

The main theorem establishing

$$P_{\max} \leq 60$$

is presented in the next section.

12 The Maximum Political Parties Theorem

We now establish the principal result of the paper.

Theorem 1 (Maximum Political Parties Theorem). *Assume*

1.

$$N_{\max} = 4,$$

2. *exactly one central bank exists within each sovereign nation,*

3. *every political party possesses exactly one domestic currency,*

4. *every political party owns shares in a non-empty subset of the four central banks,*

5. *every political party owns the unique international market portfolio,*

6. *the Injectivity Axiom holds.*

Then

$$P_{\max} \leq 60.$$

Proof. By Lemma 1,
there exist

$$4$$

possible domestic currencies.

By Lemma 2,
there exist

$$4$$

central banks.

By Lemma 3,
the number of admissible non-empty ownership subsets equals

$$2^4 - 1 = 15.$$

By Lemma 4,
the international market portfolio contributes exactly one admissible ownership state.
Consequently,
the total number of admissible ownership configurations equals

$$4 \times 15 \times 1 = 60.$$

The Injectivity Axiom states that distinct political parties possess distinct ownership configurations.
Therefore,
the number of political parties cannot exceed the number of ownership configurations.
Hence

$$P_{\max} \leq 60.$$

□

Corollary 1. *If every admissible ownership configuration is realised by exactly one political party, then*

$$P_{\max} = 60.$$

Proof. The theorem establishes

$$P_{\max} \leq 60.$$

If every admissible ownership configuration is occupied,
then all sixty ownership configurations correspond to political parties.
Therefore

$$P_{\max} = 60.$$

□

13 An Equivalent Counting Formula

The previous theorem admits a concise representation.

Proposition 1. *Under the assumptions of the model,*

$$P_{\max} = N_{\max} (2^{N_{\max}} - 1).$$

Proof. Each political party selects

1. one domestic currency,
2. one non-empty subset of central banks,
3. the unique international market portfolio.

Since

$$N_{\max} = 4,$$

the number of domestic currencies equals

$$N_{\max}.$$

The number of admissible central-bank ownership subsets equals

$$2^{N_{\max}} - 1.$$

The international market portfolio contributes

$$1.$$

Hence

$$P_{\max} = N_{\max} (2^{N_{\max}} - 1).$$

□

Corollary 2. *Substituting*

$$N_{\max} = 4$$

yields

$$P_{\max} = 4(2^4 - 1) = 60.$$

14 Computational Enumeration

The combinatorial structure naturally admits exhaustive enumeration.

The search space consists of

- four domestic currencies,
- fifteen admissible ownership subsets,
- one international market portfolio.

Consequently,

the total number of candidate ownership configurations equals

$$60.$$

The enumeration is therefore finite and complete.

15 Enumeration Algorithm

The ownership configurations may be generated by exhaustive iteration.

Pseudo-code

for currency = 1, ..., 4

for every non-empty subset S
of {B1, B2, B3, B4}

create ownership configuration

(currency, S, international portfolio)

end

end

Proposition 2. *The above algorithm enumerates every admissible ownership configuration exactly once.*

Proof. There are four outer-loop iterations.

The inner loop traverses every non-empty subset of four central banks exactly once.

Each ordered pair

$$(c, S)$$

is unique.

The international market portfolio is fixed.

Hence every ownership configuration appears once and only once.

□

16 Computational Complexity

The algorithm performs

$$4(2^4 - 1) = 60$$

iterations.

Therefore,

its time complexity is

$$O(1).$$

More generally,

for arbitrary

$$N,$$

the complexity becomes

$$O(N2^N).$$

Since

$$N_{\max} = 4,$$

the algorithm executes in constant time within the proposed architecture.

17 Graph-Theoretic Interpretation

The ownership architecture naturally induces a finite graph.

Let

$$G = (V, E),$$

where

$$V = V_N \cup V_B \cup V_I,$$

with

$$V_N = \{N_1, N_2, N_3, N_4\},$$

$$V_B = \{B_1, B_2, B_3, B_4\},$$

and

$$V_I = \{I\}.$$

The edge set is defined by

1. each nation being connected to its central bank,
2. each central bank being connected to the international market portfolio,
3. each political party selecting exactly one nation, one or more central banks, and the international portfolio.

Consequently, every political party corresponds to a connected labelled subgraph satisfying the admissibility constraints.

The ownership graph therefore contains exactly

$$60$$

admissible labelled configurations.

18 Political Economy Interpretation

Traditional political parties are frequently classified according to ideology.

Examples include

- conservative,
- liberal,
- socialist,
- nationalist,
- religious,
- environmental,
- regional,
- libertarian.

The present framework adopts an alternative viewpoint.

Political parties are classified according to institutional ownership rather than ideological preference.

Each political party becomes an economic organisation whose identity is determined by

1. its domestic monetary system,
2. its central-bank ownership structure,
3. its participation in the international market portfolio.

Ideological differences may still exist.

However, they are not part of the mathematical definition adopted in the present architecture.

19 Financial Interpretation

The ownership configuration

$$(c, S, M)$$

resembles a constrained investment portfolio.

The domestic currency provides

- transactional liquidity,
- accounting denomination,
- domestic purchasing power.

The central-bank ownership component represents exposure to the national monetary authorities.

The international market portfolio represents exposure to the global investment opportunity set.

Consequently, each political party possesses a balance sheet containing

- domestic monetary assets,
- central-bank equity,
- international financial assets.

The combinatorial theorem therefore classifies institutional balance sheets rather than electoral organisations alone.

20 Institutional Economics

Institutions coordinate economic activity by defining ownership, governance, and decision-making mechanisms.

Within the proposed architecture, political parties become institutional investors.

Their admissible ownership configurations are finite because the underlying monetary architecture is itself finite.

The theorem

$$P_{\max} \leq 60$$

therefore follows from institutional constraints rather than behavioural assumptions.

21 International Relations

The four-nation architecture implies a finite international system.

Every political party simultaneously possesses

1. domestic monetary exposure,
2. cross-border central-bank ownership,
3. participation in the international financial system.

Consequently, domestic political decisions cannot be analysed independently of international financial linkages.

The fifth currency functions as the common monetary component connecting all sovereign monetary systems.

22 Welfare Economics

Suppose

$$W_i$$

denotes a social welfare index associated with political party i .

The theorem established in the previous section does not rank the welfare levels generated by alternative parties.

Instead, it bounds the number of admissible institutional structures under the model.

Consequently,

$$P_{\max} \leq 60$$

is a structural theorem rather than a welfare theorem.

Additional assumptions concerning preferences, production, redistribution, or public goods would be required to compare welfare outcomes across ownership configurations.

23 Statistical Interpretation

The finite ownership space

$$\Omega = \{\Pi_1, \Pi_2, \dots, \Pi_{60}\}$$

may be regarded as a discrete sample space.

Let

$$X : \Omega \rightarrow \mathbb{R}$$

denote any statistic defined over ownership configurations.

Examples include

- expected monetary concentration,
- expected central-bank diversification,
- ownership entropy,
- systemic interconnectedness.

Probability distributions may therefore be defined over the finite ownership space,

allowing empirical or simulated analyses while preserving the underlying combinatorial constraints.

24 Computer Science Interpretation

The ownership configurations form a finite state space.

Each state is represented by

$$(c, S, M).$$

A convenient implementation consists of

- one integer representing the domestic currency,
- one four-bit binary mask representing central-bank ownership,
- one Boolean value representing ownership of the international market portfolio.

Since the Boolean component is fixed,

every ownership configuration may be encoded by

- two bits for the domestic currency,
- four bits for the central-bank ownership mask.

Thus every admissible ownership configuration admits a compact binary representation suitable for computational enumeration.

25 Data Science and Artificial Intelligence

The ownership configurations may serve as discrete states for computational experiments.

Possible applications include

- graph-based analysis,
- network optimisation,
- combinatorial search,
- reinforcement-learning environments,
- agent-based simulations,
- institutional stress testing.

Within such applications, the theorem restricts the admissible state space to at most

60

ownership configurations under the assumptions of the model.

26 Discussion

The principal contribution of the present paper is the identification of a finite upper bound arising from explicit institutional assumptions.

The argument is fundamentally combinatorial.

It does not rely upon

- election outcomes,
- demographic characteristics,
- ideological classifications,
- historical contingencies.

Instead, the theorem follows from

1. the four-nation architecture,
2. one central bank per nation,
3. one domestic currency per political party,
4. non-empty central-bank ownership,
5. a unique international market portfolio,
6. the Injectivity Axiom.

Together, these assumptions define a finite ownership space containing no more than sixty distinct political parties.

27 Limitations

The principal theorem of the present paper is conditional upon the architectural assumptions introduced in the preceding sections.

In particular, the following assumptions are fundamental.

1. $N_{\max} = 4$.
2. Each sovereign nation possesses exactly one central bank.
3. Every political party possesses exactly one domestic currency.
4. Every political party owns shares in a non-empty subset of central banks.
5. Every political party owns the unique international market portfolio.
6. The Injectivity Axiom holds.

If any of these assumptions are modified, the resulting combinatorial bound may change.

Accordingly, the theorem should be interpreted as a structural result within the proposed institutional architecture.

28 Generalisation

The preceding argument naturally extends to an arbitrary number of sovereign nations.

Proposition 3. *Assume*

1. *there exist N sovereign nations,*
2. *each nation possesses one central bank,*
3. *every political party possesses exactly one domestic currency,*
4. *every political party owns shares in a non-empty subset of central banks,*
5. *a unique international market portfolio exists,*
6. *the Injectivity Axiom holds.*

Then

$$P_{\max} \leq N(2^N - 1).$$

Proof. There exist

$$N$$

possible domestic currencies.

The number of non-empty subsets of N central banks equals

$$2^N - 1.$$

The international market portfolio contributes one admissible state.

Hence the number of admissible ownership configurations equals

$$N(2^N - 1).$$

The Injectivity Axiom implies that distinct political parties correspond to distinct ownership configurations.

Therefore,

$$P_{\max} \leq N(2^N - 1).$$

□

Corollary 3. *For*

$$N = 4,$$

the proposition reduces to

$$P_{\max} \leq 4(2^4 - 1) = 60.$$

29 Summary Table

Quantity	Maximum Value
Number of nations	4
Number of currencies	5
Number of central banks	4
Number of non-empty central-bank ownership sets	15
International market portfolios	1
Maximum political parties	60

Table 2: Summary of the ownership architecture.

30 Future Research

Several natural extensions remain open.

1. Relaxation of the Injectivity Axiom.
2. Dynamic formation and dissolution of political parties.
3. Weighted central-bank ownership.
4. Continuous ownership proportions.
5. Endogenous creation of international market portfolios.
6. Dynamic institutional networks.
7. Empirical calibration using observed ownership data.
8. Connections with network optimisation and institutional stability.

31 Conclusion

The paper introduced an ownership-based definition of political parties within a four-nation, five-currency institutional architecture.

Rather than defining political parties by ideology, electoral competition, or constitutional status, the proposed framework characterises each party by its ownership configuration.

Under the stated assumptions, the ownership space is finite.

The resulting combinatorial argument establishes the bound

$$P_{\max} \leq 60.$$

More generally,

$$P_{\max} \leq N(2^N - 1),$$

under the corresponding generalized assumptions.

The result illustrates how institutional constraints may induce finite combinatorial limits on organisational structures within a monetary architecture.

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Glossary

$N_{\max}$ Maximum number of sovereign nations.

$C_{\max}$ Maximum number of currencies.

$P_{\max}$ Maximum number of political parties under the model assumptions.

B_i Central bank associated with nation i .

S Non-empty subset of central banks owned by a political party.

c Domestic currency associated with a political party.

M Ownership of the international market portfolio.

Injectivity Axiom Assumption that distinct political parties have distinct ownership configurations.

Ownership Configuration The ordered triple

$$(c, S, M).$$

The End